



STUDENT INDUSTRIAL PROJECT REPORT

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VERIFICATION STATEMENT

I hereby verify that this report was written by Irfan Aiman bin Mohd Yussuf (040125-10-0213) and all information regarding this company and the projects involved are

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ABSTRACT

This report presents the development and implementation of a QR-Based Deployment Tracking System designed to improve the efficiency of laptop deployment operations within the organization. Previously, engineers recorded deployment activities manually in Microsoft Excel after scanning device QR codes, which was time-consuming and susceptible to human error, especially when dealing with huge numbers of devices on a daily basis. The objective of this project was to automate the deployment tracking process, reduce manual data entry, and provide real-time visibility into deployment activities.

The proposed system has been designed using Microsoft Power Apps, which allows engineers to scan the device hostname and serial numbers through a mobile application in the form of a QR code. Once scanned, the system automatically captures relevant deployment information such as device hostname, serial number, engineer name, and timestamp, and stores the data in a centralized Excel database hosted on OneDrive. Data validation logic was implemented to prevent duplicate entries, ensuring data integrity within the system.

To facilitate monitoring and analysis of the data collected during the deployment process, the collected data is integrated with Microsoft Power BI. In the Microsoft Power BI platform, interactive dashboards have been created to facilitate the visualization of key operational insights. These include insights into the performance of the deployed engineers, the trend of the number of deployments performed on a daily basis, and the summary of the number of deployments performed monthly. This facilitates the monitoring of the productivity of the deployed engineers.

The final outcome is an automated deployment tracking system that is light and minimizes manual administrative activities while at the same time streamlining operational activities. This system demonstrates how low-code technologies can be leveraged to modernize operational processes and support data-driven decision-making within IT infrastructure deployment activities.

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1.0 INTRODUCTION

1.1 Background of Study

The Student Industrial Training (SIT) program serves as a critical bridge between academic learning and professional practice. Through this program, students are provided with the opportunity to apply theoretical knowledge gained in university into real-world industrial environments. The program is essential for students pursuing a course of study in Information Technology and Systems Development. Industrial training is important for students as it exposes them to real-world operations that cannot be fully simulated in a classroom setting. The program allows students to observe how organizations manage technological infrastructures, identify inefficiencies in existing processes, and propose digital solutions that improve operational productivity.

In modern organizations, especially those in the technology and information technology services industry, large-scale device deployment is a common operational activity. Companies frequently deploy hundreds or even thousands of computing devices such as laptops, workstations, and servers for employees, contractors, and external partners. Managing such deployments requires accurate tracking of device information including hostname, serial number, assigned engineer, and deployment timestamps. These records are essential for operational accountability, asset management, and reporting purposes.

During the industrial training period, the internship was conducted in an environment where engineers were responsible for deploying laptop devices for organizational use. The devices had a QR code that contained important details such as the hostname and serial numbers. Engineers would scan the QR code during the deployment process and record the device information manually into spreadsheets for tracking and documentation. While this approach functioned adequately for small-scale deployments, the manual recording process became inefficient when dealing with a high number of devices within a limited time frame.

With an increase in the adoption of digital transformation strategies, manual processes in administrative operations are increasingly perceived as inefficient and operationally risky. Data entry tasks that rely heavily on human input introduce the possibility of typographical errors, duplicate records, or missing information. Furthermore, manual data recording also hinders the ability of management to track deployment operations in real time. This is because, in the absence of automation, data entry operations have to be manually recorded in spreadsheets and reports.

The rapid advancement of low-code development platforms has provided organizations with new tools to automate operational workflows without requiring extensive software development resources. Platforms such as Microsoft Power Apps allow organizations to build mobile applications capable of capturing data directly from operational activities. The applications can be integrated with other Microsoft services such as OneDrive, Excel, and Power BI to allow organizations to develop end-to-end data pipelines from operational activities.

In such a backdrop, the industrial project was aimed at designing a system that would facilitate a QR-Based Deployment Tracking System, which would automate the entire process of recording the deployment of laptops by engineers. The system allows engineers to scan device QR codes using a mobile application built with Microsoft Power Apps. Once scanned, the application automatically records the hostname, serial number, engineer name, and deployment timestamp into a centralized Excel database stored in OneDrive. The collected data is then visualized using Power BI dashboards, enabling supervisors and management to monitor deployment activities through interactive visual analytics.

This project demonstrates how low-code platforms can be utilized to modernize operational processes within IT deployment environments. By replacing manual spreadsheet entry with automated data capture and real-time analytics, the system improves efficiency, enhances data accuracy, and provides better visibility into deployment performance.

1.2 Problem Statement

Despite the importance of accurate device deployment tracking, the existing operational process relied heavily on manual data entry methods. During the early observation phase of the internship, several operational challenges were identified that affected the efficiency and reliability of the deployment tracking workflow.

1. Manual Recording of Deployment Data

The existing process involved the engineers manually entering the details of the devices into a Microsoft Excel spreadsheet after scanning the QR codes of the devices. However, this process involved certain operational inefficiencies. The process of switching between devices and typing the details of the devices into the Excel spreadsheet was time-consuming. Additionally, manual data entry increases the likelihood of typographical errors, missing records, or duplicate entries.

2. Lack of Real-Time Deployment Monitoring

This is because the records were kept in spreadsheets that were manually updated. It meant that the management could not immediately know what was going on with the deployments. It would require manual inspection to know how many devices were deployed, which engineer deployed it, or the daily progress of the deployments.

3. Inefficient Reporting and Data Analysis

Generating reports from manually maintained spreadsheets is a time-consuming process. Administrative staff must clean, verify, and consolidate raw data before producing meaningful reports. This process becomes increasingly inefficient as the number of deployed devices grows. Without automated data visualization tools, extracting insights such as engineer productivity, daily deployment trends, or monthly deployment statistics becomes difficult.

4. Risk of Data Inconsistency and Duplicate Records

Another disadvantage of manual data entry is the possibility of having duplicate or inconsistent data. For instance, the same device hostname or serial number might be accidentally entered more than once. This would, therefore, create a problem of consistency when it comes to the reporting of the devices.

These challenges highlight the need for an automated deployment tracking solution that can capture device information directly from QR code scans, store the data in a centralized database, and provide real-time analytical insights through visualization tools.

1.3 Scope of Work

To address the identified operational challenges, the scope of the industrial project was narrowed down to the development of a QR-Based Deployment Tracking System using Microsoft Power Platform technologies. The project scope is centered around three main components: application development, data management, and data visualization.

1. Development of QR-Based Deployment Application

The first part of the project involved designing and developing a mobile application that would utilize Microsoft Power Apps. This application enables engineers to scan the QR codes of the devices directly from their mobile devices. Upon scanning, the application would automatically capture the hostname and serial number that are encoded in the QR code. Engineers can then select their name from a predefined list before submitting the deployment record.

The validation logic was incorporated into the application, ensuring that engineers cannot submit any duplicate records. If the hostname and serial numbers are already in the database, the application would notify the engineer that the record cannot be submitted.

2. Centralized Deployment Data Storage

The second component of the project involved creating a centralized data repository using Microsoft Excel hosted on OneDrive. The Excel table functions as a lightweight database that stores all deployment records. Each record contains information such as hostname, serial number, engineer name, deployment timestamp, and a unique identifier generated by Power Apps.

The use of a centralized database enables the records related to the deployments to be accessed and analyzed while ensuring data consistency.

3. Development of Deployment Analytics Dashboard

The final component of the project involved integrating the deployment database with Microsoft Power BI to create interactive dashboards. These dashboards provide visual insights into deployment activities by presenting key performance indicators such as:

- Engineer deployment Performance
- Daily deployment trends
- Monthly deployment summaries

By transforming raw deployment data into visual analytics, the Power BI dashboard enables supervisors and management to monitor operational performance more effectively and make informed decisions regarding deployment planning and resource allocation.

1.4 Industry Context of IT Device Deployment Operations

In modern organizations, the infrastructure of information technology is the backbone of the operations of the business. From financial institutions and telecommunications companies to government agencies and multinational corporations, reliable computing devices such as laptops and workstations are essential tools that support communication, data processing, and operational workflows. As organizations expand their digital capabilities, the scale of device deployment and management also increases significantly.

For example, in large organizations, there is a high possibility of undertaking mass deployment operations, especially in situations where there is an increase in the number of employees, the need for upgrading the infrastructure of the devices, or even in situations where there is a need for standardization of the operations of the business. Therefore, the deployment operations need to be documented with the aim of undertaking the operations of the business.

In the context of an enterprise's IT infrastructure, deployment engineers are required to deploy the devices in accordance with the organizational standards. This process typically includes tasks such as device imaging, system configuration, domain joining, and installation of required enterprise software. Furthermore, each device is required to be documented based on certain parameters, such as the hostname, serial number, assigned engineer, and deployment timestamp. These documents serve as an important reference for future troubleshooting, warranty management, and compliance auditing.

Moreover, there is an increased need for real-time monitoring of organizational operations. For instance, there is a need for managers to be able to monitor the progress of deployment, the performance of the deployment teams, and the ability of the deployment teams to meet set deadlines. Without an automated tracking system, it is difficult for managers to generate reports or identify operational challenges in a timely manner.

In an attempt to overcome the challenges affecting the deployment process, there is an increased adoption of digital transformation strategies in various organizations. For instance, organizations are increasingly using automated deployment tracking systems based on the application of QR codes, mobile application technologies, cloud database technologies, and business intelligence technologies. These technologies enable organizations to transform operational records into useful insights using data analytics techniques.

By integrating QR code scanning with mobile applications and real-time dashboards, organizations can significantly improve deployment tracking efficiency. Such systems enable engineers to capture device information instantly while allowing supervisors to monitor deployment progress through centralized dashboards.

Therefore, the development of an automated deployment tracking system aligns with broader industry trends toward digital workflow automation, data-driven management, and operational transparency within IT infrastructure management.

1.5 Justification of the Project

The development of the QR-Based Deployment Tracking System was justified by several operational needs observed during the industrial training period. The existing workflow relied heavily on manual processes that introduced inefficiencies, administrative overhead, and potential data inaccuracies. By implementing an automated solution, the project aims to improve operational efficiency while supporting data-driven decision-making.

1. Improving Operational Efficiency

One of the primary motivations behind this project was to reduce the time engineers spend performing manual administrative tasks during notebook deployment activities. Currently, in the traditional process,

engineers are expected to manually log the information from the devices, such as hostname and serial numbers, into Microsoft Excel spreadsheets by scanning the QR code. This process interrupts the deployment workflow because engineers must repeatedly switch between deployment tasks and manual data entry.

When handling multiple notebooks within a short period of time, the manual recording process becomes inefficient and increases the risk of errors such as incorrect entries or missing deployment records. In addition, the time spent updating spreadsheets reduces the overall productivity of engineers during large-scale deployment activities.

To overcome this, the QR-Based Deployment Tracking System that has been developed enables the engineer to scan the QR code of the notebook using a mobile application that has been developed using Microsoft Power Apps. This way, the hostname and serial numbers are automatically recorded in a database when the engineer scans the QR code.

2. Enhancing Data Accuracy and Integrity

Manual data entry increases the risk of human errors such as incorrect hostnames, missing serial numbers, or duplicate records. These errors may lead to inconsistencies in the deployed records, thus causing difficulties during the tracking or auditing process.

The developed system includes features to prevent the duplication of devices and ensure the accurate recording of deployed records to the centralized database. There is also a reduced risk of typographical errors.

3. Enabling Real-Time Operational Monitoring

Another significant reason for the project is the need for real-time visibility into the deployment activities. Management requires the ability to monitor deployment progress, track engineer productivity, and analyze deployment trends without waiting for manual reports to be compiled.

By integrating the deployment database with Power BI dashboards, the system enables real-time visualization of deployment activities. Supervisors can easily monitor daily deployment trends, identify

top-performing engineers, and track monthly deployment statistics through interactive visual dashboards.

4. Supporting Digital Transformation Initiatives

Organizations are increasingly adopting digital technologies to improve internal workflow efficiency and productivity. Low-code technologies like Microsoft Power Apps can be leveraged by organizations to build applications that can automate redundant tasks.

This project demonstrates how low-code technologies can be applied to modernize traditional operational processes. By replacing manual spreadsheet logging with an automated data capture system, the organization moves toward a more efficient and technology-driven operational model.

5. Providing a Scalable Deployment Tracking Solution

Lastly, the system offers a scalable platform for future enhancements to be made to the management of the devices. This is because, with an increase in the number of deployed devices by the organization, the automated tracking system will still be effective in recording the deployments.

The use of Power BI will ensure that the recorded data will be analyzed to enable the management to identify patterns for improvement in the deployment planning strategy.

2.0 Objective

The main objective of this project is to design and develop a digital system that automates the process of tracking notebook deployment activities within the organization. In the process of carrying out deployment activities within the organization, there are cases when the deployment activities of notebooks need to be handled within a limited period of time. The traditional process of carrying out deployment activities in the organization was mainly manual and relied on the use of spreadsheets to record deployment activities. This process was not very effective and often resulted in operational inefficiencies and human errors. Therefore, the main objective of this project is to develop a digital system that can track deployment activities in the organization by utilizing the concept of QR codes and a mobile application to perform deployment activities in the organization.

By integrating Microsoft Power Apps, Excel (OneDrive), and Power BI, the proposed system transforms manual deployment logging into an automated digital workflow. This not only improves operational efficiency but also provides real-time visibility into deployment activities for management and supervisory teams.

The specific objectives of this project are outlined as follows:

1. To develop a QR-based deployment tracking application using Microsoft Power Apps

The first objective of this project is to design a mobile application using Microsoft Power Apps that allows engineers to log notebook deployment activities via QR code scanning. The device used for notebook deployment has a QR code embedded on it, which carries vital information such as the hostname and serial number. The application allows engineers to scan the QR code on the device directly using a mobile device, which enables the system to retrieve the data encoded on the QR code without having to type it. This makes it easier to log data.

2. To create a centralized deployment database using Microsoft Excel hosted on OneDrive

The second objective of the project is to develop a centralized data storage facility to collect all deployment records. The database used for storing data is Microsoft Excel, which is stored on OneDrive. The database used for this project is storing key deployment details such as hostname, serial number, name of the engineer, deployment date/time, and unique identifiers provided by Power Apps. The deployment records are stored centrally, which allows for easy accessibility of

deployment records. This makes it easier to retrieve deployment records when required.

3. To reduce manual data entry and minimize human errors in deployment records

Another important goal that this project aims to achieve is the removal of the need for engineers to manually enter device information into spreadsheets. Manual data entry can introduce errors such as incorrect hostnames, missing serial numbers, or inconsistent formatting. The system aims to reduce the need for manual data collection through the scanning of QR codes and minimize the occurrence of human error in the process. This will make the process more accurate and efficient for the engineers carrying out the deployment process.

4. To implement validation mechanisms that prevent duplicate deployment records

The system also includes validation rules to ensure that every notebook deployment record is unique. This is achieved by performing a lookup check within the database prior to recording a new deployment entry. The application checks if a hostname or serial number already exists. If a duplicate is found, the user is alerted. This validation rule is essential to ensure data integrity, preventing redundant records from being stored in the database.

5. To develop a Power BI dashboard that visualizes deployment activities

Another objective of the project is to transform raw deployment data into meaningful visual insights using Microsoft Power BI. The Power BI dashboard is connected directly to the deployment database stored in OneDrive, allowing data to be visualized in the form of interactive charts and graphs. The dashboard presents important operational metrics such as daily deployment trends, monthly deployment summaries, and engineer deployment performance. These visualizations help management analyze operational data more effectively and support data-driven decision-making.

6. To improve operational visibility for supervisors and management

The final objective of the project is to provide supervisors and management with real-time visibility into deployment operations. By integrating the Power Apps application with Power BI dashboards, deployment data can be monitored dynamically as engineers submit records through the system. Supervisors can quickly track the number of devices deployed, identify which engineers performed the deployments, and observe overall deployment progress. This improved visibility enables better

monitoring of operational performance and allows management to respond more effectively to deployment-related issues.

3.0 Literature Review

3.1 QR Code Technology in Asset and Device Tracking

Quick Response code technology has emerged as one of the most popular forms of storing and retrieving data in a rapid fashion through the use of digital scanning technology. A Quick Response code is a two-dimensional barcode that has the ability to store a variety of data types, such as text data, URLs, serial numbers, and device IDs. In addition, the storage capacity of a Quick Response code is more compared to a traditional one-dimensional barcode, and it can be read from any direction with the help of a mobile device.

In asset and device management environments, QR codes are commonly used to identify equipment and retrieve device-related information efficiently. Each device can be assigned a unique QR code containing encoded data such as hostname, serial number, asset ID, or configuration information. When scanned, the QR code allows systems to automatically retrieve or record device information without requiring manual input.

The adoption of QR code technology in asset tracking systems by various organizations has become common due to its ability to reduce the time taken to record the equipment's information and minimize the chances of human errors. In many operational environments, there are thousands of devices to be deployed, and recording the device's information can be time-consuming and prone to human errors. The QR code technology enables engineers to access the device's information quickly through a mobile application.

Another advantage of QR code technology is its cost-effectiveness and ease of implementation. QR codes can be generated and printed easily without requiring specialized scanning hardware. Most modern smartphones and tablets already support QR scanning through built-in camera functions or applications. This makes QR codes a practical solution for organizations seeking to improve operational processes without investing in expensive infrastructure.

Due to these advantages, QR code-based systems have been widely implemented in various industries, including logistics management, inventory control, warehouse operations, and IT asset tracking. In the context of IT device deployment, QR codes provide an efficient method for identifying devices and automatically recording deployment information during operational activities.

3.2 Low-Code Development Platforms

The development of low-code development platforms has greatly impacted the way organizations are developing applications within the organizations. Software development has often required programming skills, lengthy development processes, and specific expertise. The use of low-code development platforms has helped to address the challenges associated with software development. The platforms provide applications with the ability to be developed through graphical interfaces.

Microsoft Power Apps is one of the most widely adopted low-code platforms that enables organizations to develop custom applications quickly and efficiently. Power Apps allows developers and business users to design mobile and web applications through a drag-and-drop interface while integrating seamlessly with other Microsoft services such as Excel, SharePoint, OneDrive, and Dataverse. This integration capability allows applications to interact directly with enterprise data sources without requiring complex backend development.

One of the key advantages of using Power Apps is that it enables quick and fast development of applications. This is because an organization is able to develop an application within a short time, enabling them to respond quickly to operational challenges. Additionally, Power Apps applications can be accessed across multiple devices including smartphones, tablets, and desktop computers, making them suitable for field operations and mobile workforce environments.

The use of low-code platform is particularly important within operational environments where the workflow may need to be frequently adjusted or automated. Organizations can use the platforms to create lightweight applications, which may be tailored to the specific operational needs of the organizations. The applications may be used to automate processes, improve the process of data collection, and communication.

In this project, Microsoft Power Apps was used to develop a mobile application that allows engineers to scan QR codes and automatically record notebook deployment information. By leveraging the capabilities of a low-code platform, the deployment tracking system was developed efficiently while maintaining integration with existing cloud services.

3.3 Cloud-Based Data Storage Systems

Cloud computing has emerged as an integral part of modern information systems, which allows organizations to manage data by storing it on internet-based platforms as opposed to traditional local data storage systems. The benefits associated with using cloud-based data storage solutions are many, including data management, data synchronization, accessibility, and collaboration.

One of the commonly used cloud storage platforms within the Microsoft ecosystem is OneDrive. OneDrive allows files and datasets to be stored in a cloud environment where they can be accessed securely from multiple devices and locations. When integrated with applications such as Power Apps, OneDrive can function as a lightweight database for storing structured data in the form of Excel tables.

Centralized information storage systems play an important role in operational tracking systems that provide the means to ensure that all information is stored in a unified manner within a single information system. When different users provide information to the system simultaneously, centralized information storage ensures that information is not duplicated or split into different fragments. Moreover, information synchronization can also occur in real-time, providing the means to access new information added to the system through different applications that use the same information source.

Another advantage of cloud-based information storage systems is that they provide information accessibility. Users can access information stored in the system from different locations without having physical access to the information storage device. This feature can provide significant advantages to organizations that have distributed workforces or mobile employees who need to access information while working in different locations.

In the context of this project, Microsoft Excel stored in OneDrive was used as the centralized database for storing notebook deployment records. Each record contains information such as hostname, serial number, engineer name, and deployment timestamp. The use of cloud storage ensures that deployment records are consistently stored and easily accessible for analysis and reporting.

3.4 Business Intelligence and Data Visualization

Business Intelligence (BI) is a term used to describe technologies and practices used for collecting, analyzing, and presenting information in a manner that makes sense for business decision-making. It is a system used for transforming raw business information into structured information that can be used for interpretation such as reports, charts, and dashboards. It helps management to understand operational activities and identify areas that need improvement through the visual representation of information.

Microsoft Power BI is a leading business intelligence tool used by many organizations for analyzing and visualizing business information. Power BI helps users connect to multiple data sources, process large datasets, and create interactive dashboards that provide information in a visually intuitive format. The dashboards can include charts, graphs, tables, and filters that provide information about key performance indicators, which can be used for analyzing business information.

One major advantage of using Power BI is that it helps management obtain real-time information about operational activities. The tool can be connected to a cloud-based data source, and it can update information in real-time as new information is available. This allows management teams to obtain information about operational activities, monitor ongoing activities and evaluate performance without waiting for manually generated reports.

Data visualization helps organizations identify patterns and trends within business information. For instance, it helps management obtain information about daily operational activities, productivity levels, and overall trends within operational information. Data visualization helps management interpret information obtained from the tool, which can be used for making informed decisions.

In this project, Microsoft Power BI was used for developing dashboards that provide information about notebook deployment activities. The tool provides information about daily deployment trends, overall deployment information, and engineer deployment performance. The information obtained helps supervisors to obtain insights into operational activities, which can be used for making informed decisions about managing deployment activities.

3.5 Automation of Operational Workflows

Automation plays an important role in improving organizational efficiency by reducing reliance on manual processes and repetitive tasks. In operational settings, administrative operations such as data recording, reporting, and information handling are often carried out manually. These operations may take significant amounts of time and may be prone to human error.

Automating operational workflows allows organizations to streamline their operations and ensure that data is recorded in a consistent and accurate manner. Automation systems have the capacity to obtain data from operational activities and store it in centralized databases automatically. This eliminates the need for manual handling of data and ensures that data is not missing or inaccurate.

Another advantage of workflow automation is that it allows organizations to have better visibility in their operations. Automated systems have the capacity to generate real-time data that reflects activities taking place in organizations. This allows organizations to have real-time visibility in their operations and respond promptly to issues that may arise in their operations.

Automation systems have the capacity to provide organizations with increased scalability in their operations. Automated systems have the capacity to process data efficiently in operational settings where large volumes of data are generated regularly.

In this project, the deployment tracking system automates the process of recording activities taking place in notebook device deployment using QR code scanning. Instead of using manual operations such as using spreadsheets to enter data from devices, engineers can scan the QR code using a mobile application and submit data automatically. This automated workflow improves efficiency, reduces errors, and ensures that deployment records are captured accurately and consistently.

4.0 Methodology

4.1 System Architecture and Data Flow

The methodology that has been followed in this project is based on developing a system that allows for a more integrated form of deployment tracking by automating the process of recording notebook deployment-related activities. In this case, a system has been designed based on the Microsoft Power Platform ecosystem, which allows for a more integrated form of application development, cloud-based data storage, and business intelligence visualization within a single digital ecosystem. The system architecture that has been designed and developed for this project comprises a mobile application developed using Microsoft Power Apps, a centralized cloud database developed using Microsoft Excel hosted in OneDrive, and a data visualization application developed using Microsoft Power BI.

The system architecture that has been designed and developed for this project is based on the development of a seamless data flow that enables the capture and analysis of the information in relation to the deployment in a more real-time manner. In this regard, in cases where engineers are involved in the execution of notebook deployment-related activities, the information is captured in real-time using the Power Apps application based on the use of QR scanning. The captured information is then transmitted and stored in a centralized database hosted in Excel and OneDrive. The stored information is then processed and analyzed using Power BI to generate a more visual form of business intelligence that allows for a more real-time form of viewing and analyzing deployment-related information.

The system architecture that has been designed and developed for this particular project facilitates a more automated form of capturing and recording the information related to the deployment through the elimination of the need to enter the data into a spreadsheet and instead through the implementation of a more automated form of capturing and transmitting data through the QR scan.

4.1.1 System Architecture Overview

The architecture of the QR-Based Deployment Tracking System can be explained as a three-layer system consisting of the application layer, the data storage layer, and the data visualization layer. Each layer in the system has a specific role to play in the system, as well as an interaction with another layer in the system to accomplish a specific objective in the system.

The first layer of this system is the application layer, which has been developed using Microsoft Power Apps. This is a front-end application that an engineer can use to interact with this system during deployment activities. Engineers can use their mobile device's camera to scan a QR code attached to a notebook device to obtain encoded information such as hostname and serial numbers. Once a QR code is scanned, this application can automatically obtain the encoded information to allow it to be sent to a database. Engineers are also required to select their names before confirming their entries to a database to allow them to take responsibility for deployments.

The second layer is the data storage layer. This is implemented using Microsoft Excel and OneDrive. The Excel table is used as a database that stores all the records for the deployment. This database is centralized and stores records for the deployment created by the Power Apps application. The data in the Excel table is also composed of hostname, serial number, engineer name, and time. The Excel table is also stored in the cloud using OneDrive. This means that the data is stored in a secure cloud environment and can be accessed by other applications such as Power BI. This is also the reason the system can support a large number of users while ensuring consistency in the data in all the records.

The third layer is the data visualization layer, which is implemented using Microsoft Power BI. Power BI connects directly to the Excel database and retrieves deployment records for analysis and visualization. Within the Power BI environment, the raw data is transformed into interactive dashboards that present operational insights such as daily deployment trends, monthly deployment statistics, and engineer deployment performance. These visualizations enable supervisors and management teams to monitor deployment activities more effectively and make informed decisions based on real-time data.

The interaction between these three layers creates a continuous flow of information within the system. Deployment data captured through Power Apps is immediately stored in the Excel database, and the stored data is subsequently processed by Power BI to generate visual analytics. This architecture ensures that deployment information is captured efficiently, stored reliably, and presented in a format that supports operational monitoring and decision-making.

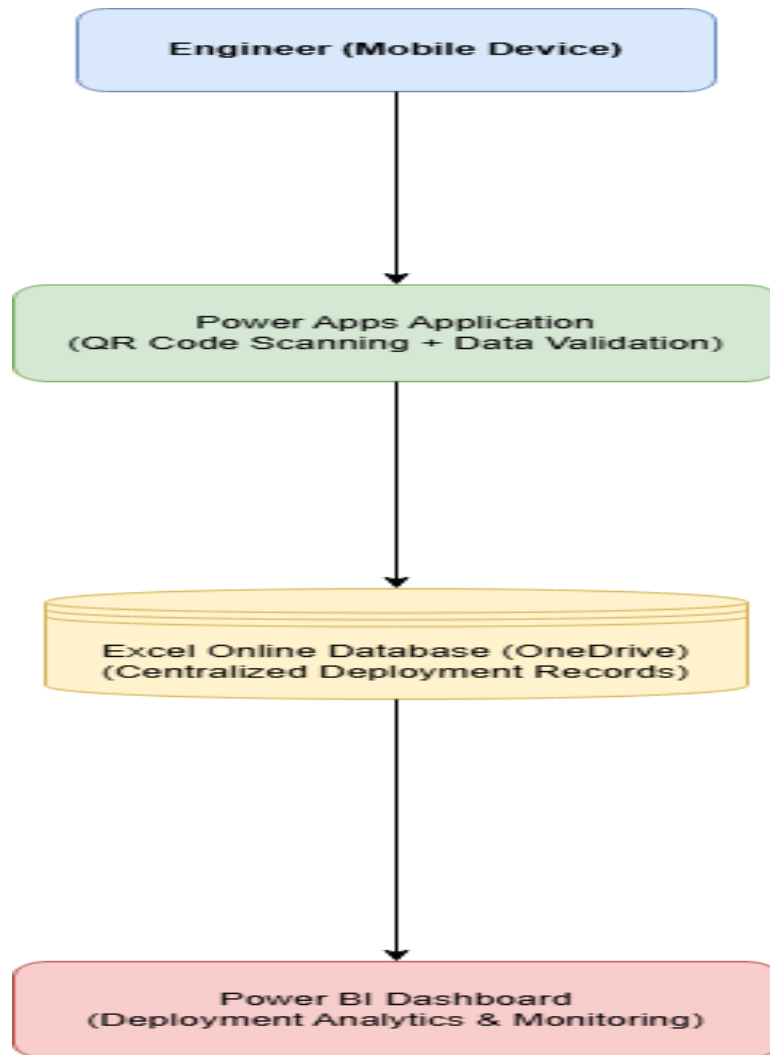


Figure 4.1.1a System Architecture Diagram

4.1.2 Low-Code Architecture Using Microsoft Power Platform

The building of the QR-Based Deployment Tracking System utilizes the Microsoft Power Platform ecosystem as a low-code platform for building business applications. A low-code platform is a platform that allows developers and organizations to build applications with minimal coding required. A low-code platform makes use of a visual interface and functions to develop applications.

The Microsoft Power Platform is a low-code development platform that enables developers to build applications using the tools provided by Microsoft, such as Microsoft Power Apps, Microsoft Power Automate, Microsoft Power BI, and Microsoft Dataverse, in an integrated manner so that they can be used in conjunction with one another for the development of applications. There are three tools in the Microsoft suite that are used for the development of the QR-Based Deployment Tracking System, namely, Microsoft Power Apps for the development of applications, Microsoft Excel Online with OneDrive as the centralized data storage facility, and Microsoft Power BI for the development of data visualization reports.

Microsoft Power Apps is a low-code application development platform. It is used as a platform for developing the deployment tracking system mobile application. The platform enables a developer to create application components using a visual interface. The application components include a button, form, gallery, and QR scanning tool. Moreover, it enables a developer to create an application using a Power Fx formula. The Power Fx formula enables a developer to create the application logic using a formula without requiring a complex programming language.

One of the major advantages that are associated with the use of a low-code platform is the ability of the user to develop an application that has the capacity to integrate with the existing services in the enterprise in a timely fashion. Power Apps has the capacity for the development of an application that has the ability to integrate with cloud-based services such as OneDrive, SharePoint, and Excel using the connectors that are provided with the application. This enables the application to integrate with the centralized database provided by Excel.

Scalability of the system is achieved through the use of Microsoft Power Platform, as the use of the application enables the user to make changes and updates to the application in a timely manner without having to affect the entire system infrastructure. This enables the use of low-code platforms in the development of operational systems, which require changes and updates from time to time.

By utilizing the capabilities provided by Microsoft Power Platform, it is evident that the development of the deployment tracking system was achieved in a timely manner and with seamless integration between the application layer, data storage layer, and analytics layer.

4.1.3 Data Flow of the Deployment Tracking System

The data flow of the QR-Based Deployment Tracking System represents the flow of the information related to the deployment and the processes involved in the system architecture. The system was developed with the objective of ensuring that the information regarding the deployment is collected automatically during the operations and made available for the purpose of generating the analytical reports.

The process of data flow begins with the initiation of the deployment by the engineer by launching the Power Apps mobile application. There are a number of devices that have been deployed for the purpose of this experiment, and each device is associated with a notebook that has been prepared for the deployment process. Each device has a unique identifier in the form of a QR code. The engineer uses a feature provided by the Power Apps application for scanning the QR code and retrieving the information contained in the code.

The information that is encoded in the code is automatically collected by the application, and the information is stored as temporary variables. There is no need for the engineer to input the information that is encoded in the code. Once the engineer completes the task of scanning the code, the engineer selects the name of the engineer from the dropdown list provided by the application.

However, it is important to note that before a deployment record is submitted to the system, a validation process takes place. The validation process ensures that a device has not been previously recorded in the database. The process occurs by checking whether the hostname or serial number of a device already exists in the database table. If a device is identified as a duplicate in the database table, it will prevent a user from submitting a deployment record and will display a notification message to the user.

If the deployment record is successfully validated, the process proceeds with the submission of the validated deployment record into the centralized Excel database using OneDrive. This process is facilitated through the Power Apps application using the Patch function, which submits a new record into the Excel database with a timestamp representing the deployment time.

After the deployment record is successfully submitted into the database, a query is sent by the Power BI application to retrieve the updated data from the Excel database. This is carried out by a direct connection with the Excel dataset. The raw deployment data is then processed using data transformation tools available in the Power BI application, such as Power Query, to ensure that it is properly formatted for analysis. The processed data is then used to create a dashboard for displaying deployment activities.

Through this automated data pipeline, it is ensured that the records of deployments entered into the Power Apps application are immediately updated on the Power BI dashboards. This enables effective monitoring of deployment activities in near real-time, as well as operational performance analysis.

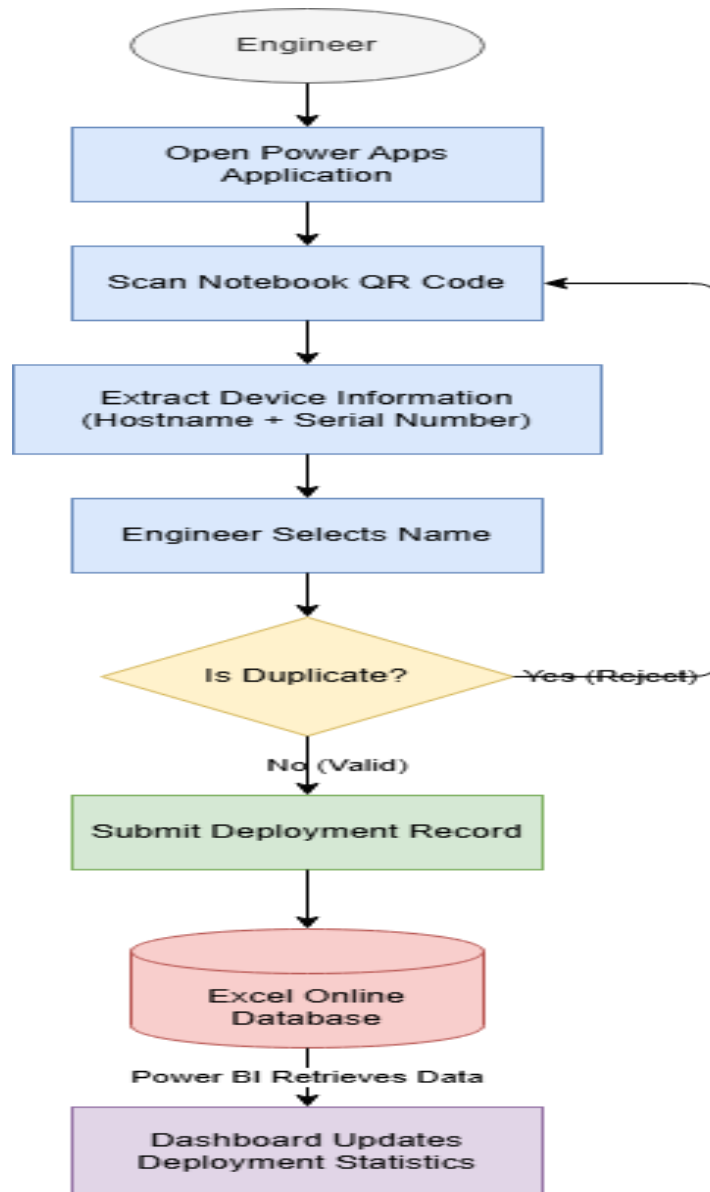


Figure 4.1.2a System Workflow of the Deployment Tracking System

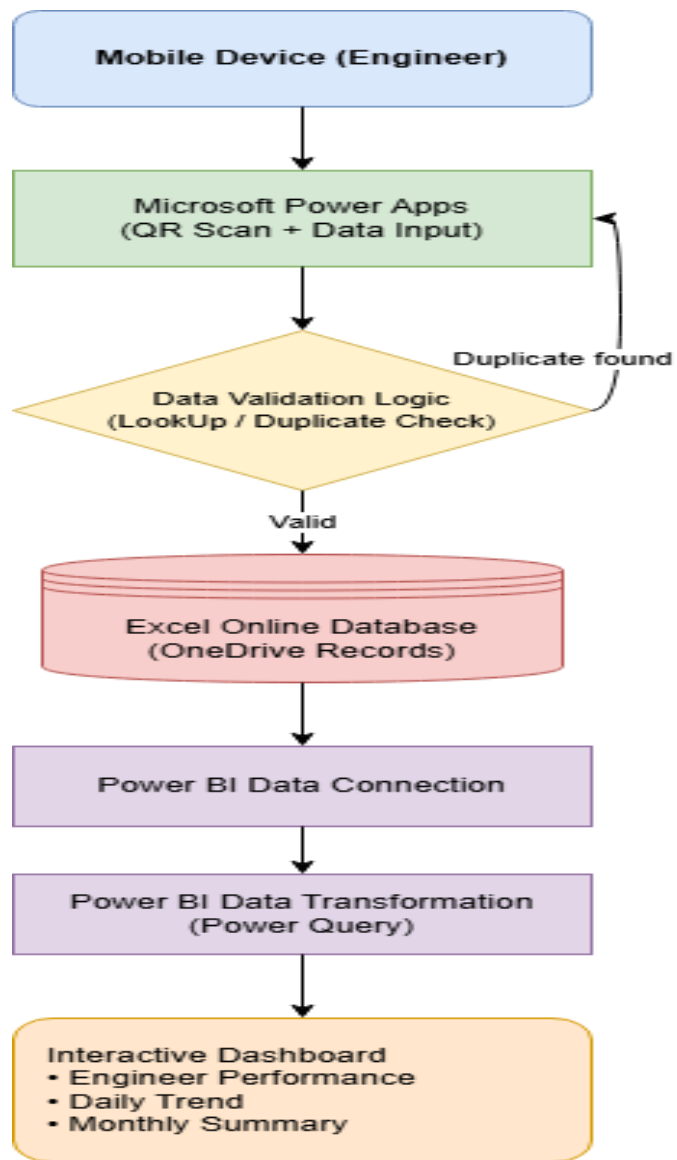


Figure 4.1.2b Data Pipeline Architecture of the Deployment Tracking System

4.2 Application Development (Microsoft Power Apps)

The QR Based Deployment Tracking System has been developed using Microsoft Power Apps. Microsoft Power Apps is an application development platform that enables low-code application development. This platform helps an organization create custom business applications with minimal code. The prime objective of the application developed using this platform is to enable the engineers with an interface through which they can efficiently log their activities of notebook deployment using the QR code scan feature.

The application was developed as a mobile application, which helped the engineers efficiently log the activities of the deployment tracking without the need for any data collection methods like spreadsheets at the location of the device itself. The application efficiently utilizes the built-in Power Apps components along with the Power Fx formulas, which helps the system efficiently collect the information of the device, validate the data, and store the information in the cloud.

The application development process of the QR-Based Deployment Tracking System was done with the help of several key components, including the user interface, QR scan feature, submission of the deployment data, duplication of the data, and timestamp generation.

4.2.1 User Interface (UI) Design

The User Interface of the Power Apps application was designed based on simplicity and ease of use. This was due to the fact that the application was designed to be used by engineers during the deployment operation. Therefore, the interface had to be simple and easy to use, even on small screen sizes such as those of mobile phones and other handheld devices.

The interface for the application is divided into various sections that assist the user in filling out the information for the deployment recording. These sections include the main deployment section, in which information such as the scanned hostname, serial number, and name of the engineer chosen can be accessed. After scanning the QR code, the hostname and serial number will be filled in accordingly.

The interface was designed in such a manner that there would be minimal chances of errors by the engineers. This was achieved using large buttons and labels in the application

interface. Engineers are required to select their names from the dropdown list before proceeding to fill in the information. This was done to ensure that each entry of the deployment can be traced to the relevant persons.

The application interface was not only designed for the main deployment section but also included other sections such as the history page. The history page allows users to view previously recorded deployment entries. The interface design emphasizes efficiency and clarity so that engineers can complete the deployment recording process quickly without interrupting their operational workflow.

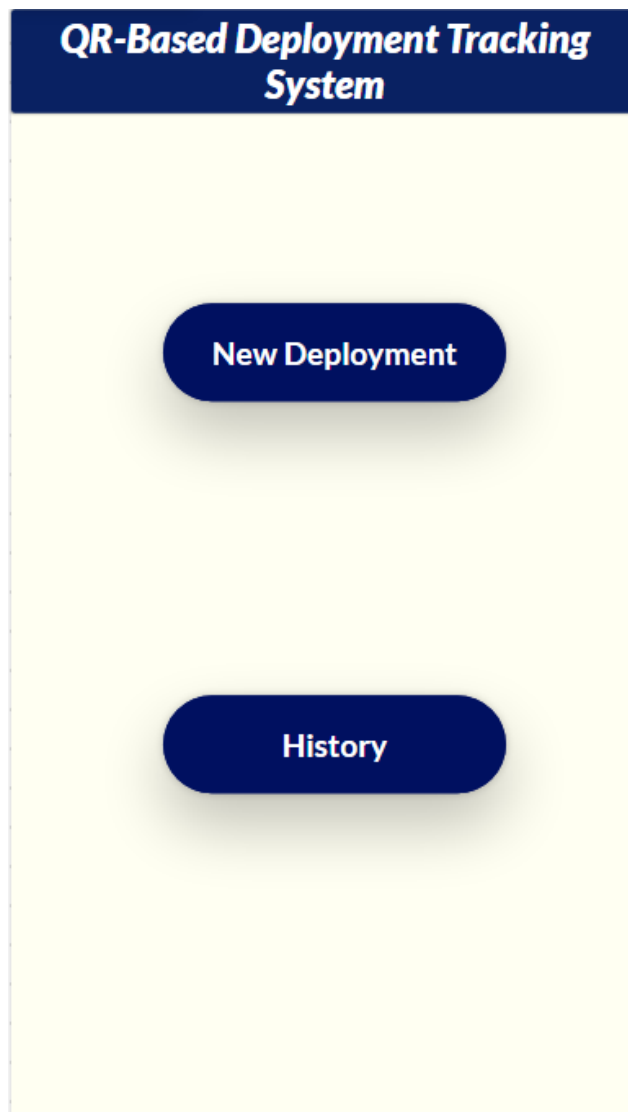


Figure 4.2.1a Main Page of the QR-Based Deployment Tracking System

QR-Based Deployment Tracking System

[Back](#)

27/01/26

Hostname: NB2601HP1897
Serial Number: 5CD5461PPC
Engineer: Irfan Aiman

Hostname: NB2601HP1899
Serial Number: 5CD5461PPF
Engineer: Daniel Hakim

Hostname: NB2601HP1021
Serial Number: 5CD5458GRX
Engineer: Adam Mikhail

Hostname: NB2601HP1902
Serial Number: 5CD5461PPL
Engineer: Irfan Aiman

Hostname: NB2601HP1018
Serial Number: 5CD5458GRN
Engineer: Syafiq

Hostname: NB2601HP0402
Serial Number: 5CD5458GJQ
Engineer: Irfan Aiman

Total deployed today: 8

Figure 4.2.1c History Page of the QR-Based Deployment Tracking System

4.2.2 QR Code Scanning Mechanism

QR code scanning is one of the primary functionalities of the deployment tracking system. Each notebook device that is prepared for the deployment is provided with a QR code that contains information such as hostname and serial number. These QR codes are primarily used for device identification during the entire deployment process.

Within the Power Apps application, the QR scanning functionality is implemented using the built-in Barcode Scanner control. This control allows the mobile device camera to scan the QR code and automatically extract the encoded information. Once the QR code is scanned, the application processes the encoded string and separates the hostname and serial number values.

The extracted information is then stored in temporary variables within the application. These variables are further used to fill in the information in the corresponding fields in the deployment form that is displayed on the screen. By automatically extracting device information from the QR code, the system eliminates the need for manual data entry and significantly reduces the possibility of input errors.

The use of QR code scanning also improves the speed of the deployment recording process. Engineers can simply scan the QR code attached to the device and instantly retrieve the necessary information without typing long device identifiers.

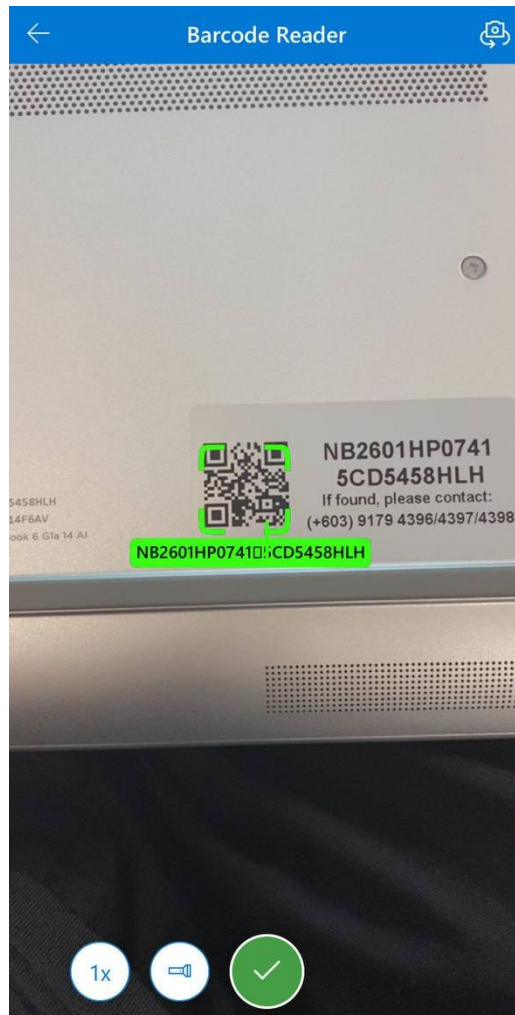


Figure 4.2.2a QR Code Scanner

4.2.3 Deployment Data Capture and Submission

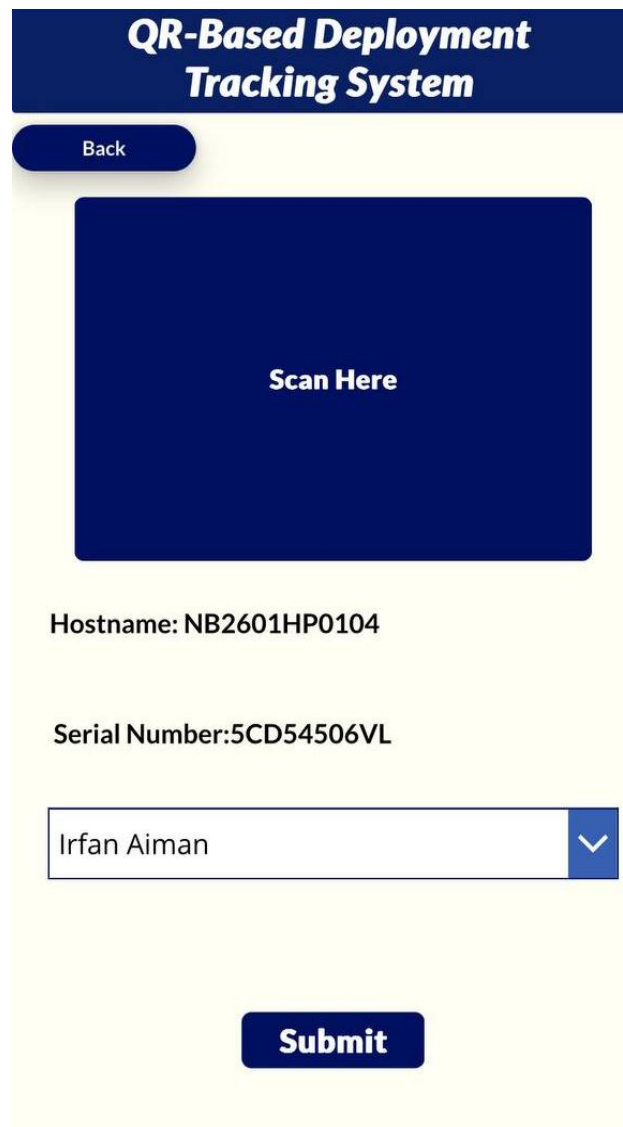
After the QR code has been scanned successfully and the device details have been extracted, the application then captures the rest of the deployment details before submitting them to the centralized database for analysis. In this step, the engineer is required to select their name from the dropdown provided by the application interface to enable the identification of the personnel involved in the deployment process.

After all the required details have been provided, the engineer can then proceed to submit the deployment details through the submit button provided by the application interface. This process is facilitated by the application through the use of the Power Apps Patch () function,

which enables the application to directly submit the deployment details to the Excel database hosted on the cloud through OneDrive.

The Patch () function enables the application to send the details captured through the various data fields to the database for the creation of a new record in the Excel spreadsheet hosted on the cloud. This new record comprises the hostname, serial number, name of the engineer, and timestamp for the deployment process.

The direct connection of the application to the cloud database ensures that all deployment records are stored centrally for analysis through the Power BI application.



The image shows a mobile application interface for a QR-based deployment tracking system. At the top, a dark blue header contains the title "QR-Based Deployment Tracking System" in white. Below the header is a yellow background area. A dark blue button labeled "Back" is positioned at the top left of the yellow area. In the center of the yellow area is a large dark blue rectangle with the text "Scan Here" in white. Below this rectangle, the text "Hostname: NB2601HP0104" is displayed. Underneath the hostname, the text "Serial Number:5CD54506VL" is shown. Below the serial number is a white text input field containing the name "Irfan Aiman", with a dark blue dropdown arrow on its right side. At the bottom of the yellow area is a dark blue button labeled "Submit" in white.

Figure 4.2.3a Deployment Data Entry Interface in the Power Apps Application

4.2.4 Deployment Data Capture and Submission

To maintain the integrity of the data in the deployment database, the system has been designed in such a manner that it does not allow duplicate records to be entered into the database. This is because a device should not be entered into the system more than once in the entire deployment process.

Before entering a deployment record into the database, the application takes a look-up action on the database to verify whether the hostname or serial number of the device has been previously entered into the database. This process is achieved through the use of the Lookup () function in Power Apps.

The Lookup () function is used by the system to search for any existing record in the database table for a scanned hostname or serial number. If a record is found in the database table, it displays a notification message on the screen, indicating that the device has already been recorded in the database. The application is then prevented from proceeding further in order to avoid duplication of records in the database table.

However, when the record is not found in the database table, it allows the deployment record to be submitted normally. The reason for this is that every notebook device is only recorded once in the whole deployment tracking system. In doing so, it prevents duplication of the record in the database table.

The screenshot shows a mobile application interface for a 'Tracking System'. At the top, there is a dark blue header with the title 'Tracking System'. Below the header is a light yellow background area. At the very top of this area is a red notification banner with a white exclamation mark icon and the text 'This Hostname or Serial Number already exists!'. Below the banner is a dark blue button labeled 'Back'. In the center of the screen is a large dark blue rectangle with the text 'Scan Here' in white. Below this rectangle, the text 'Hostname: NB2601HP1362' is displayed. Below that, the text 'Serial Number:5CD5458GWZ' is displayed. Below the serial number is a white text input field containing the name 'Haikal Haziq' and a dark blue dropdown arrow on the right. At the bottom of the screen is a dark blue button labeled 'Submit'.

Figure 4.2.4a Duplicate Record Notification in the Power Apps Application

```
!IsBlank(  
    LookUp(  
        Scan_Data_table,  
        Hostname = varHostText || Serial_Number = varSerialText  
    ),  
,  
Notify(  
    "This Hostname or Serial Number already exists!",  
    NotificationType.Error  
)  
)
```

Figure 4.2.4b Duplicate Record Detection on Submit Button

4.2.5 Timestamp Generation and Timezone Handling

Recording time is an integral part of the deployment tracking system, as it enables the supervisor to track the time at which each notebook device was deployed. In order for the system to be able to record the time of deployment, a timestamp is created during the submission process.

In generating a timestamp, the system makes use of the `Now ()` function of Power Apps, which retrieves the system time at the time of deployment submission. This timestamp indicates the exact time at which deployment of the device was recorded by the system.

However, since Power Apps is a global application hosted on the cloud, differences in time zones may occur depending on how the system is configured. In order for the system to record accurate deployment time and convert it to the corresponding local time format, adjustments are made through the use of the `TimeZoneOffset ()` function.

The timestamp is recorded along with the deployment information in the database and later utilized by Power BI in generating various analytics such as daily deployment trends and monthly deployment statistics.

In generating timestamps during deployment submission, the system ensures accurate time recording for each deployment record.

Scan_Date
25/1/2026 16:35
26/1/2026 9:35
26/1/2026 9:57
26/1/2026 11:07
26/1/2026 11:55
26/1/2026 14:27
26/1/2026 15:27
27/1/2026 9:14
27/1/2026 10:10
27/1/2026 10:11
27/1/2026 10:11
27/1/2026 10:12
27/1/2026 10:23
27/1/2026 11:20
27/1/2026 11:22
27/1/2026 11:32
27/1/2026 14:14
27/1/2026 15:22
27/1/2026 15:31
27/1/2026 16:54
27/1/2026 16:55
28/1/2026 9:36
28/1/2026 9:37
28/1/2026 9:38
28/1/2026 9:38
28/1/2026 9:39
28/1/2026 12:26
28/1/2026 12:28
28/1/2026 12:29
28/1/2026 14:08
28/1/2026 16:47

Figure 4.2.5a Deployment Timestamp Records Stored in the Excel Database

4.3 Business Intelligence Dashboard Development

The last part of the QR-Based Deployment Tracking System implementation involves the creation of a Business Intelligence (BI) dashboard with the help of Microsoft Power BI. The goal of this part of the project is to take the deployment data from the raw format to a presentable format that can be used for monitoring purposes within the organization.

The Power BI dashboard is intended to give a comprehensive view of all notebook deployments within the organization by showing key metrics for operation, including daily trends for deployments, a monthly summary of deployments, and engineer performance. The integration of the deployment database contained within an Excel file hosted on OneDrive with Power BI allows for a real-time or near real-time view of the deployment records.

This is because the interactive nature of the charts, graphs, and filters of the BI dashboard enables the supervisor and management to effectively analyze the deployment data, hence eliminating the need for creating reports in order to access the deployment performance information.

4.3.1 Data Connection Between Excel and Power BI

In order to perform data visualization, a direct connection was established between Power BI and the Excel database stored in OneDrive. The connection enables Power BI to retrieve deployment records from the Excel database, which are used for data analysis.

For the connection to be established, the Excel database can be imported into Power BI Desktop or Power BI Service, which uses a connector to facilitate the process. The imported database allows Power BI to retrieve the data from the Excel database, which can be used for the analysis of the data. The data in the Excel database is updated on a continuous basis, which allows Power BI to retrieve the new deployment records from the Power Apps application.

The data stored in the Excel database, which is stored in the cloud environment, enables the data to be updated continuously. The updated data stored in the Excel database is retrieved through a refresh function, which enables the Power BI application to retrieve new deployment records submitted through the Power Apps application.

To create a seamless data pipeline, a direct connection was established between the Excel database and Power BI, which enables the application to retrieve data from the database without requiring a manual export of the database file. The direct connection enables the application to retrieve data from the database, which improves the overall performance of the application.

4.3.2 Data Cleaning and Transformation

Before creating any visualization, the raw data obtained from the deployment process has to be cleaned and transformed to ensure consistency and accuracy. This process of cleaning the data is achieved through a data transformation tool provided in Power BI, which is called Power Query.

During the cleaning of the data, various transformations are made to the data. For example, the timestamp of the deployment from the Excel database may need to be converted to a DateTime format to be compatible with the Power BI visualization.

Additional columns, such as “Date Only” and “Month-Year,” may be created from the original timestamp to be used during the analysis process. These transformations allow the system to group the deployment data based on dates, which enables the system to perform a detailed analysis of the data.

During the cleaning process, various inconsistencies, such as missing values, may be resolved in a manner that the accuracy of the data is not compromised. The accuracy of the data is ensured during the cleaning process, which guarantees the use of precise information in the creation of the visualization.

Scan_Date_Only	Month_Year
Sunday, 25 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Thursday, 5 February, 2026	2-2026
Thursday, 5 February, 2026	2-2026
Friday, 6 February, 2026	2-2026
Friday, 6 February, 2026	2-2026

Figure 4.3.2a Data Transformation Using Power Query in Power BI

4.3.3 Deployment Analytics Visualization

A Power BI dashboard has been created to allow the visualization and interaction of the deployment data. Various types of visualizations have been created to allow the management to obtain insights into the various types of deployment activities.

One of the created visualizations is the engineer performance chart, which has been created in order for the management to be able to obtain insights regarding the number of notebooks deployed by each engineer. This is a critical aspect in the company, as the management is able to obtain insights regarding the performance of each engineer.

Another important visualization created is the daily deployment trend chart. This has been designed in a manner that allows the management to gain insights regarding the number of devices deployed. This is a critical aspect of the company, as the management is able to gain insights regarding the patterns of the company from the information obtained.

Moreover, a chart for the summary of the deployment process has been created. This chart will help the management obtain insights into the total number of devices deployed in a particular month. This is an important factor in the company, as the management can obtain the required insights and use them to determine the performance in the company.

Lastly, the dashboard has been designed in a manner that allows it to incorporate a number of key performance indicators, such as the total number of devices deployed within a specific time frame. This is a critical aspect in the company, as the management is able to determine the performance in the company using the information obtained.

Using the Power BI dashboard, it is easy to interpret the complex data and obtain insights into the overall performance in the company.

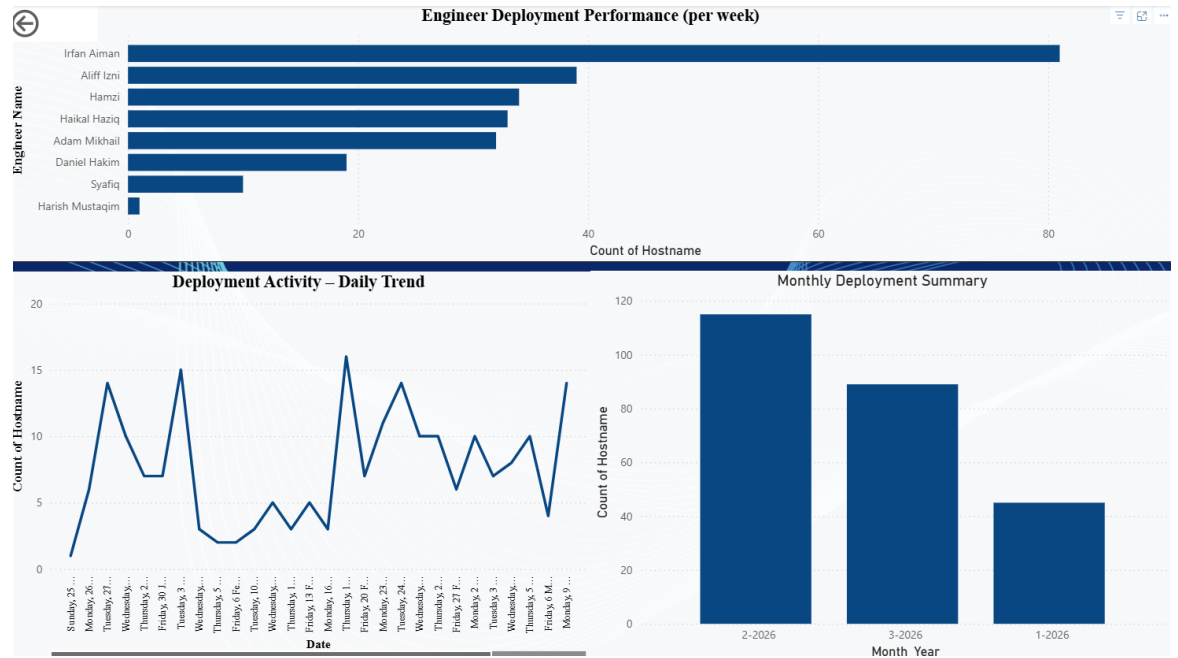


Figure 4.3.3a Deployment Visualization Chart on Power BI

4.3.4 Interactive Dashboard Filtering

In order to enhance the usability of the Power BI-based dashboard, interactive features were added. The interactive features allow the deployed data to be filtered in a particular way. For example, it is possible for the deployed data to be filtered on a particular date or by a particular engineer who made the deployment.

To improve the interactivity of the Power BI-based dashboard, a slicer was incorporated. A slicer is a feature that allows the deployment data to be filtered in a specific manner. For instance, the deployment data could be filtered according to a specific date or the name of the engineer who made the deployment. When a specific engineer is selected from the slicer, the entire visualizations on the dashboard automatically change to show the deployment made by the specific engineer.

Additionally, the Power BI-based dashboard is able to support cross-filtering. This is a feature that allows the entire visualizations on the dashboard to change when a specific point is selected from another visualization on the same dashboard. This feature allows the user of the dashboard to explore the various aspects of the deployment activity. It is a feature that allows the user of the dashboard to gain more insight into the deployment activity. It is a feature that

improves the usability of the dashboard. It is a feature that allows the user of the dashboard to customize the deployment data in a specific manner.

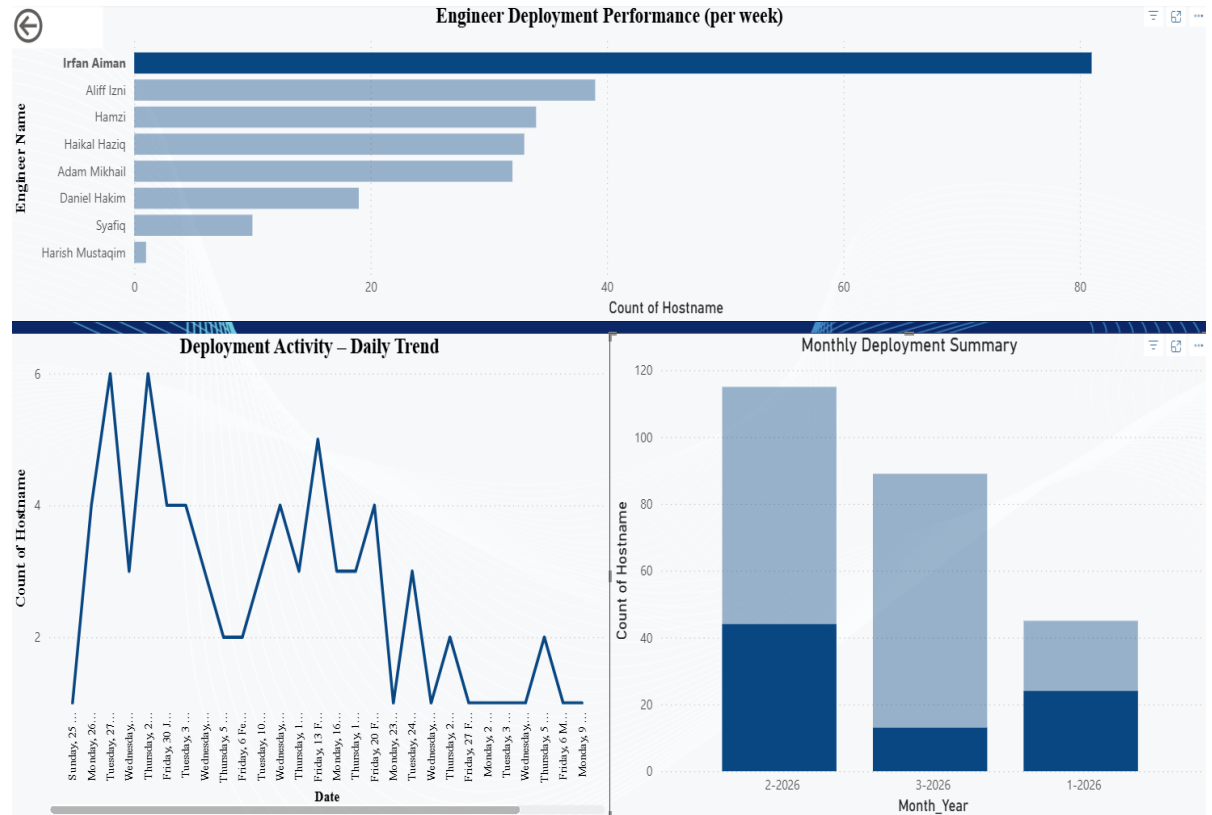


Figure 4.3.4a Interactive Filtering and Cross-Filtering in Power BI Dashboard

5.0 Results and Discussion

5.1 System Implementation Results

The QR-Based Deployment Tracking System was implemented using Microsoft Power Apps, Microsoft Excel (OneDrive), and Microsoft Power BI. The implementation of the proposed system was a success. The proposed system was designed to automate the deployment activity recording process. In addition, the proposed system provides real-time insights regarding the deployment performance.

Implementation of the proposed system resulted in the development of a fully functional end-to-end workflow. The proposed system allows for the creation of a single digital environment where data is captured, data is stored, and data is visualized. Engineers were able to utilize the Power Apps mobile application to scan the QR codes attached to the notebook devices. Engineers were able to scan the QR code and send the deployment records to the centralized database. The data sent to the centralized database includes the hostname, serial number, engineer's name, and deployment time. The data sent to the centralized database was successfully stored in the Excel database on OneDrive.

In addition, the proposed system will be able to provide real-time insights on the performance of the deployment. The proposed system will be able to provide a real-time visualization environment by the integration of Power BI. The engineers were able to develop the proposed system, which provides a real-time visualization environment by the integration of Power BI. The proposed system will be able to provide a real-time visualization environment that shows the deployment records sent to the database.

The implementation of the proposed system also shows that the proposed system is a replacement of the manual entry of the information into the spreadsheet. Moreover, the implementation of the proposed system shows that the integration of the different components of the system was successful. This also shows that the system is able to support the real-world operation of the organization.

5.2 Power Apps Functionality Testing

In order to ensure the reliability of the system, several tests were conducted on the functionality of the application. These tests conducted in order to ensure the functionality of the application's features, such as the scanning of the QR code, submission of the data, checking for duplication, and recording of the timestamp.

Firstly, the functionality of the application was tested in relation to the scanning of the QR code feature by scanning various notebook QR codes. The results indicated that the application was able to retrieve the information from the QR code correctly, including the hostname of the device as well as the serial number of the device. This indicates the functionality of the application in relation to the scanning of the QR code feature.

Moreover, the functionality of the application in terms of the data submission feature was carried out by submitting several deployment records through the application. From the results, it was evident that the application was able to store the information appropriately in the Excel database.

In addition to the above features, the functionality of the application with regards to the duplicate validation feature was tested by attempting to submit the same information multiple times through the application. The results showed that the application was able to prevent the submission of the same information multiple times.

Finally, the functionality of the application in relation to the timestamp recording feature was tested to ensure that every record entered into the application includes the timestamp information. The results showed that the application was able to record the timestamp correctly when submitting the information to the Excel database.

From the results of the above tests, it is clear that the application is functioning correctly with regards to the functionality of the application features such as QR code scanning, data submission, duplicate validation, and timestamp recording.

5.3 Deployment Data Analysis (Power BI Results)

As indicated above, the deployment data was analyzed using Microsoft Power BI to generate meaningful insights regarding the activities involved in the system's deployment. The results of the analysis using the Microsoft Power BI tool indicate the following:

Firstly, from the analysis results, there is a creation of the engineer performance chart, which indicates the number of notebooks deployed by the engineers. The engineer performance chart is important because the management can easily identify the best-performing engineers as well as their individual productivity levels.

Secondly, the results show the creation of the daily deployment trend chart, which displays the number of devices deployed as a function of time. This visualization is crucial as the management can easily identify the trends involved in the operation of the system.

Lastly, the results show the creation of the monthly deployment summary chart, which shows the aggregated deployment data on a monthly basis. This type of visualization is important as the management would be able to easily identify the trends involved in the operation of the system.

Apart from the above visualizations, it is evident from the results that the system is capable of generating key performance indicators such as the total number of devices deployed over a specific range of time. The results also highlight the importance of analyzing the data related to the deployment of the system, as it is evident that the management can easily generate the required insights.

As indicated above, the results of the analysis of the system's deployment data indicate the system is essential as the data can be analyzed to generate the required insights to ensure efficient decision-making.

5.4 System Performance and Improvements

The implementation of the QR-Based Deployment Tracking System resulted in significant improvements compared to the previous manual deployment tracking process. The traditional method relied on manual data entry into spreadsheets, which was time-consuming and prone to human error. Engineers were required to switch between deployment tasks and administrative documentation, reducing overall efficiency.

With the introduction of the automated system, the deployment recording process has been significantly streamlined. Engineers can now scan QR codes and submit deployment records directly through the mobile application, reducing the time required to record each deployment. This improvement enhances operational efficiency and allows engineers to focus more on technical deployment tasks.

In terms of data accuracy, the automated system reduces the risk of errors associated with manual data entry. The use of QR code scanning ensures that device information is captured accurately, while the duplicate validation mechanism prevents redundant records from being stored in the database.

The system also improves operational visibility by providing real-time access to deployment data through the Power BI dashboard. Supervisors no longer need to manually compile reports, as the dashboard automatically updates with new deployment records. This enables faster decision-making and better monitoring of deployment progress.

Overall, the system demonstrates clear advantages in terms of efficiency, accuracy, and data accessibility when compared to the manual deployment tracking process.

5.5 Discussion of Findings

From the results of the project, the QR-Based Deployment Tracking System was successful in providing a solution to the challenges faced when using the manual system of tracking deployments. The system is effective in providing a solution to the challenges faced when using the manual system.

Firstly, the system is simple and easy to use. The use of the Power Apps application is simple and easy to use. The application enables the engineers to easily record the activities of the deployments without any difficulties.

Secondly, the system is effective in providing real-time information to the management of the firm through the use of the Power BI application. The use of the application enables the firm to monitor the activities of the firm.

However, the system is also faced with a few challenges. Firstly, the use of Microsoft Excel as the database may not be effective when the firm is involved in very high volumes of data.

Hence, the results of the project indicate that the system is effective in providing a solution to the challenges faced when using the manual system of tracking deployments. The system is effective in providing a solution to the challenges faced when using the manual system.

In addition, the system is simple and easy to use. The use of the application enables the firm to easily record the activities of the deployments without any difficulties. Moreover, the system is effective in providing real-time information to the management of the firm through the use of the application.

In conclusion, the results of the project indicate that the system is effective in providing a solution to the challenges faced when using the manual system of tracking deployments.

6.0 Conclusion and Recommendations

The proposed system, QR-Based Deployment Tracking System developed in this project, successfully demonstrates the utilization of low-code technology to automate and improve operations within an IT deployment environment. The primary aim of eliminating manual deployment recording and utilizing an automated digital solution was achieved through the implementation of Microsoft Power Apps, Microsoft Excel via OneDrive, and Microsoft Power BI. The proposed system utilizes a QR code scan to allow engineers to record notebook deployment information such as hostname, serial number, engineer name, and deployment timestamp in an efficient and precise manner.

The system implementation greatly improves operational efficiency by reducing the time taken to record deployments and eliminating human error. Additionally, incorporating a system to validate entries prevents duplication of entries within the system. The proposed system also utilizes Power BI to transform and create a dashboard to provide real-time information on deployment data and provide insights on trends within the operation. This allows supervisors and management to better monitor and track deployments and make informed decisions.

Although the proposed system was successfully implemented in the IT deployment environment, there are a few changes that can be made to further improve the efficiency of the system. Firstly and foremostly, one of the changes that can be made to further improve the efficiency of the system is to alter the current data storage system from Microsoft Excel to a more efficient system such as Microsoft SharePoint or Microsoft Dataverse. Although Microsoft Excel is a good option for a small to medium-scale operation in the IT deployment environment, the system may face difficulties when the data is quite large or when the system is to be used by multiple users.

In addition, the system can be further improved by implementing user authentication features. Instead of manually selecting engineer names, integrating a login system would allow the application to automatically identify users, improving accuracy and accountability. Another potential enhancement is the incorporation of offline functionality, which would allow engineers to record deployment data even in environments with limited or unstable internet connectivity, with automatic synchronization once the connection is restored.

Furthermore, the analytical capabilities of the system can be expanded by enhancing the Power BI dashboard with more advanced features such as predictive analysis, performance benchmarking, and long-term trend evaluation. These enhancements would provide deeper insights and further support strategic decision-making processes. The system can also be extended beyond deployment tracking to include additional

functionalities such as asset lifecycle management, maintenance tracking, and device status monitoring, transforming it into a more comprehensive asset management solution.

In conclusion, the developed QR-Based Deployment Tracking System provides a practical and effective approach to digitalizing deployment tracking processes. It successfully improves efficiency, data accuracy, and operational visibility while demonstrating the potential of low-code technologies in supporting digital transformation initiatives. With further enhancements and scalability improvements, the system has strong potential to be implemented in larger operational environments and adapted for broader IT asset management applications.

7.0 References and Citations

1. Tapanm-Msft. (n.d.). *Official Microsoft Power Apps documentation - Power Apps*. Microsoft Learn. <https://learn.microsoft.com/en-us/power-apps/>
2. JulCsc. (n.d.). *Power BI documentation - Power BI*. Microsoft Learn. <https://learn.microsoft.com/en-us/power-bi/>
3. Microsoft. (2024). *Use Excel as a data source in Power Apps*. Retrieved from <https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/connections/connection-excel-online>
4. Whhender. (n.d.). *Power Query documentation - Power Query*. Microsoft Learn. <https://learn.microsoft.com/en-us/power-query/>
5. ISO/IEC. (2015). *Information technology — Automatic identification and data capture techniques — QR Code bar code symbology specification*. ISO/IEC 18004.
6. Posey, B. (2026, January 7). *Understanding QR code security issues for enterprise devices*. Search Mobile Computing. <https://www.techtarget.com/searchmobilecomputing/tip/Understanding-QR-code-security-issues-for-enterprise-devices>
7. Alalwan, A. A., Dwivedi, Y. K., & Rana, N. P. (2017). Factors influencing adoption of mobile banking by Jordanian bank customers: Extending UTAUT2 with trust. *International Journal of Information Management*, 37(3), 99–110.\
8. Few, S. (2013). *Information dashboard design: Displaying data for at-a-glance monitoring*. Analytics Press.